

Performance Testing

Date	04 july 2026
Team ID	xxxxxxx
Project Name	Human Development Index(HDI) Predictor
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1 : Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	HDI Dashboard & Data Processing
Test Environment	Windows 11, Python 3.x, Streamlit
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario/ Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load dashboard	20	60	Loads within 2 sec
2	Country-wise analysis	30	60	Correct results
3	Generate charts	25	60	Charts render
4	Apply filters	40	60	Quick response

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status(Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Within limit
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Acceptable
3	Throughput (Req/sec)	≥ 100 req/sec	120 req/sec	Pass	Stable
4	Error Rate	< 1%	0.2%	Pass	Low errors
5	CPU Utilization	< 80%	65%	Pass	Normal
6	Memory Utilization	< 80%	58%	Pass	Stable

Step 4: Observations & Analysis

PROJECT TITLE / NAME: <u>Human Development Index (HDI) Prediction System</u>		
KEY FINDINGS	BOTTLENECKS IDENTIFIED	OPTIMIZATION STEPS TAKEN
- The Human Development Index prediction model successfully predicted HDI values based on input indicators such as life expectancy, education, and income. - The model achieved good prediction accuracy after data preprocessing and feature selection. - The web application displayed prediction results quickly and correctly for user inputs.	- Some datasets contained missing or inconsistent values, requiring preprocessing. - Prediction accuracy was slightly lower for countries with limited historical data. - Initial model training took longer due to the size of the dataset.	- Cleaned the dataset by removing missing values and duplicates. - Applied feature scaling and selected the most important features to improve model performance. - Tuned machine learning model parameters to increase prediction accuracy. - Optimized the web application to reduce prediction response time and improve user experience.

Step5: Screenshots/Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

